

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2020 P1HR Worked Solutions

Jan 2020 | P1HR

27 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Ratio Problem Solving**

Jan 2020 P1HR - Jan 2020 | P1HR

- 1** Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6
The **total** amount of money that Asha and Julie receive is \$36

Work out the amount of money that Brendon receives.

\$.....

(Total for Question 1 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Ratio Problem Solving**

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WORKED SOLUTION

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 3****TOPIC: Fractions**

Jan 2020 P1HR - Jan 2020 | P1HR

2 Show that $3\frac{1}{5} \times 2\frac{5}{8} = 8\frac{2}{5}$

(Total for Question 2 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 2

Marks: 3

TOPIC: **Fractions**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3** **Marks: 8****TOPIC: Rearranging Formulas**

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- 3** (a) Make a the subject of $d = g + 2ac$

.....
(2)

- (b) Factorise fully $9ef - 12f$

.....
(2)

- (c) Expand and simplify $(x + 2)(x - 5)$

.....
(2)

- (d) Simplify fully $\frac{n^4 \times n^7}{n^5}$

.....
(2)

(Total for Question 3 is 8 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 8

TOPIC: **Rearranging Formulas**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Set Notation & Venn Diagrams**

Jan 2020 P1HR - Jan 2020 | P1HR

4 $B = \{b, l, u, e\}$

$G = \{g, r, e, y\}$

$W = \{w, h, i, t, e\}$

(a) List all the members of the set

(i) $B \cup G$

(ii) $W \cap G'$

.....
(2)

Serena writes down the statement $B \cap G \cap W = \emptyset$

(b) Is Serena's statement correct?

You must give a reason for your answer.

.....
(1)

.....
(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 3

TOPIC: **Set Notation & Venn Diagrams**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5** **Marks: 3****TOPIC: Area & Perimeter**

Jan 2020 P1HR - Jan 2020 | P1HR

- 5 The diagram shows Yuen's garden.

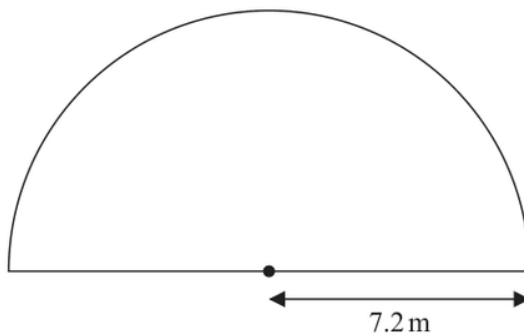


Diagram **NOT**
accurately drawn

The garden is in the shape of a semicircle of radius 7.2 m.

Yuen is going to cover his garden with grass seed.

Yuen has 12 boxes of grass seed.

Each box of grass seed contains enough seed to cover 6 m^2 of the garden.

Has Yuen enough grass seed for his garden?

Show your working clearly.

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 3

TOPIC: **Area & Perimeter**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Solving Quadratic Equations**

Jan 2020 P1HR - Jan 2020 | P1HR

- 6 Solve $x^2 - 5x - 36 = 0$
Show clear algebraic working.

(Total for Question 6 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 3

TOPIC: **Solving Quadratic Equations**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 3**TOPIC: **Percentages**

Jan 2020 P1HR - Jan 2020 | P1HR

- 7 In a sale, the normal price of a hat is reduced by 15%
The sale price of the hat is 20.40 euros.

Work out the normal price of the hat.

..... euros

(Total for Question 7 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: Percentages

Jan 2020 P1HR - Jan 2020 | P1HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Statistics Toolkit**

Jan 2020 P1HR - Jan 2020 | P1HR

- 8 5 children are playing on a trampoline.
The mean weight of the 5 children is 28 kg.

2 of the children get off the trampoline.
The mean weight of these 2 children is 26.5 kg.

Work out the mean weight of the 3 children who remain on the trampoline.

..... kg

(Total for Question 8 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 8

Marks: 3

TOPIC: **Statistics Toolkit**

Jan 2020 P1HR - Jan 2020 | P1HR

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9** **Marks: 5****TOPIC: Standard & Compound Units**

Jan 2020 P1HR - Jan 2020 | P1HR

9 Pablo made a solid gold statue.

He melted down some gold blocks and used the gold to make the statue.
Each block of gold was a cuboid, as shown below.

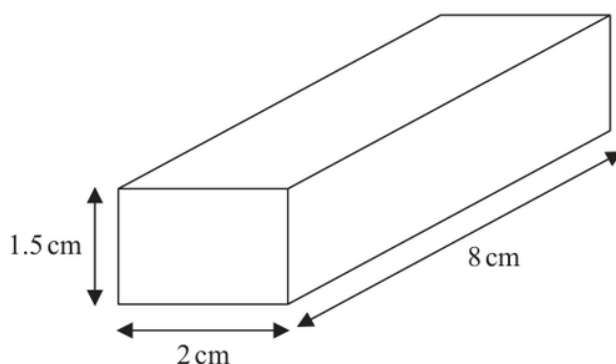


Diagram **NOT**
accurately drawn

The mass of the statue is 5.73 kg.
The density of gold is 19.32 g/cm^3

Work out the least number of gold blocks Pablo melted down in order to make the statue.
Show your working clearly.

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 5

TOPIC: **Standard & Compound Units**

Jan 2020 P1HR - Jan 2020 | P1HR

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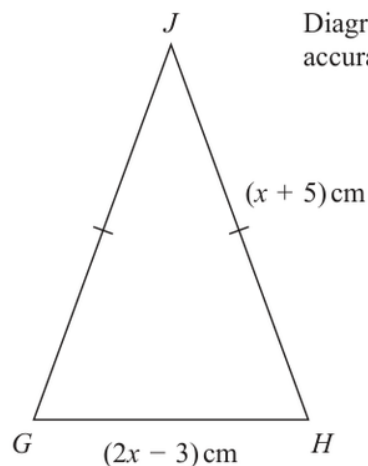
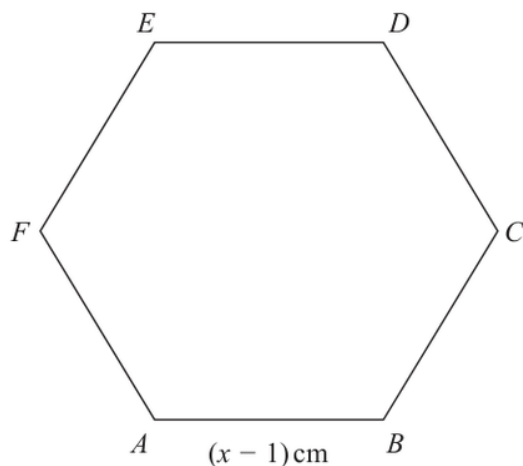
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 10** **Marks: 5****TOPIC:** Area & Perimeter

Jan 2020 P1HR - Jan 2020 | P1HR

- 10** The diagram shows a regular hexagon, $ABCDEF$, and an isosceles triangle, GHJ .

Diagram **NOT**
accurately drawn

The perimeter of the hexagon is equal to the perimeter of the triangle.

Find the length of each side of the hexagon.

Show clear algebraic working.

..... cm

(Total for Question 10 is 5 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 5

TOPIC: **Area & Perimeter**

Jan 2020 P1HR - Jan 2020 | P1HR

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jan 2020 P1HR - Jan 2020 | P1HR

11 The weight of a cat is 4.3 kg correct to 2 significant figures.

(a) Write down the upper bound of the weight of the cat.

..... kg
(1)

(b) Write down the lower bound of the weight of the cat.

..... kg
(1)

$$G = e - f$$

 $e = 17$ correct to the nearest integer $f = 9.4$ correct to one decimal place(c) Work out the upper bound for the value of G .

.....
(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

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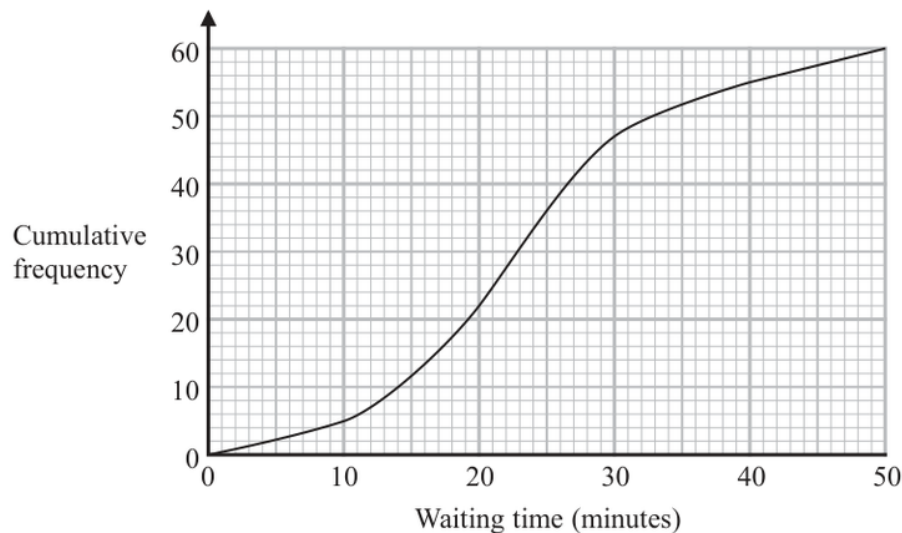
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 12****Marks: 5****TOPIC: Cumulative Frequency Diagrams**

Jan 2020 P1HR - Jan 2020 | P1HR

- 12** The cumulative frequency graph gives information about the waiting times, in minutes, of people with appointments at Hospital A.



- (a) Use the graph to find an estimate of the median waiting time at Hospital A.

..... minutes
(1)

- (b) Use the graph to find an estimate of the interquartile range of the waiting times at Hospital A.

..... minutes
(2)

At a different hospital, Hospital B, the median waiting time is 28 minutes and the interquartile range of the waiting times is 19 minutes.

- (c) Compare the waiting times at Hospital A with the waiting times at Hospital B.

.....
(2)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 5

TOPIC: **Cumulative Frequency Diagrams**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 4****TOPIC: Algebraic Proof**

Jan 2020 P1HR - Jan 2020 | P1HR

13 (a) Use algebra to show that $0.\dot{5}\dot{7}\dot{2} = \frac{63}{110}$

(2)

Given that y is a prime number,

(b) express $\frac{3}{2 - \sqrt{y}}$ in the form $\frac{a + b\sqrt{y}}{c - y}$ where a , b and c are integers.

(2)

(Total for Question 13 is 4 marks)

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PAST PAPER SOLUTION

Q#: 13

Marks: 4

TOPIC: **Algebraic Proof**

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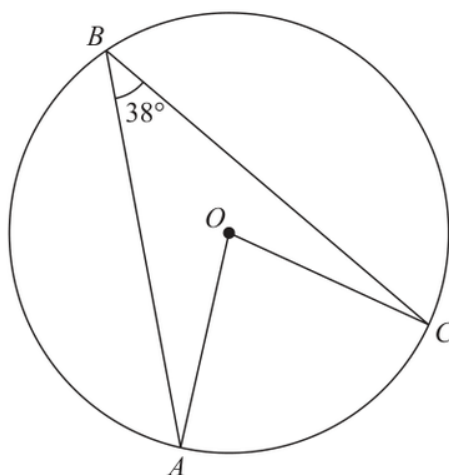
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 4****TOPIC: Circle Theorems**

Jan 2020 P1HR - Jan 2020 | P1HR

14Diagram **NOT**
accurately drawn

A , B and C are points on a circle, centre O .

Angle $ABC = 38^\circ$

Work out the size of angle OAC .

Give a reason for each stage of your working.

o

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14

Marks: 4

TOPIC: **Circle Theorems**

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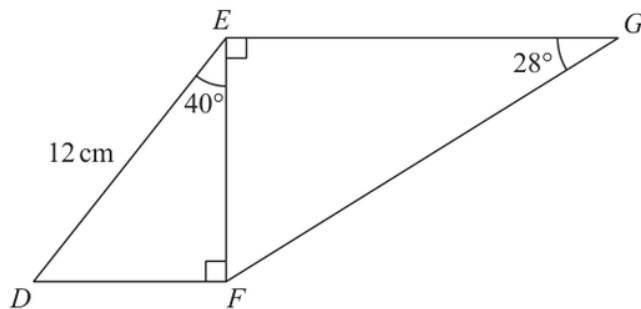
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EA

PAST PAPER QUESTION**Q#: 15** **Marks: 4****TOPIC:** Right-Angled Triangles - Pythagoras & Trigonometry

Jan 2020 P1HR - Jan 2020 | P1HR

15 The diagram shows two right-angled triangles, DEF and EFG .Diagram **NOT**
accurately drawnWork out the length of EG .

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Combined & Conditional Probability**

Jan 2020 P1HR - Jan 2020 | P1HR

16 Steffi is going to play one game of tennis and one game of chess.

The probability that she will win the game of tennis is 0.6

The probability that she will win **both** games is 0.42

Work out the probability that she will **not** win either game.

(Total for Question 16 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 16

Marks: 4

TOPIC: **Combined & Conditional Probability**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Functions**

Jan 2020 P1HR - Jan 2020 | P1HR

17 The function f is such that $f(x) = (x - 4)^2$ for all values of x .

(a) Find $f(1)$

.....
(1)

(b) State the range of the function f .

.....
(1)

The function g is such that $g(x) = \frac{4}{x+3}$ $x \neq -3$

(c) Work out $fg(2)$

.....
(2)

.....
(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17

Marks: 4

TOPIC: **Functions**

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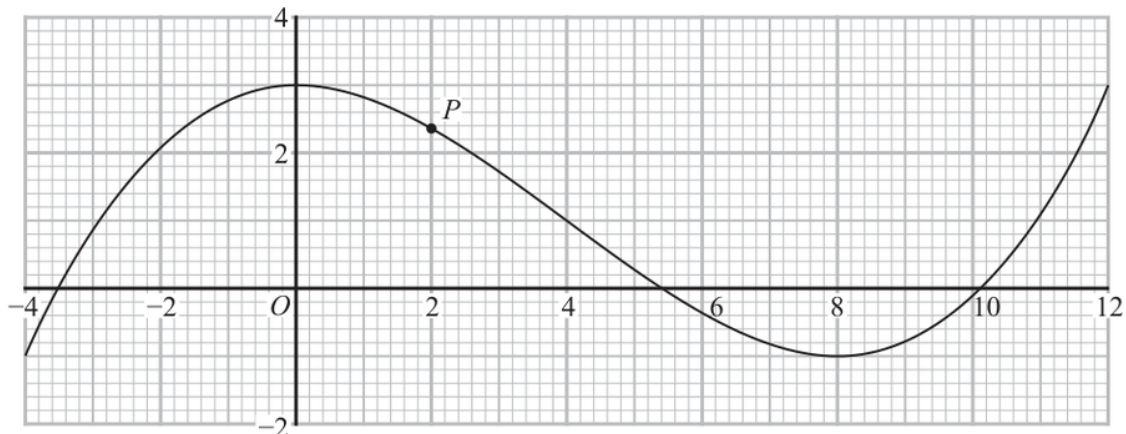
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18** **Marks: 7****TOPIC: Estimating Gradients**

Jan 2020 P1HR - Jan 2020 | P1HR

18 The diagram shows the graph of $y = f(x)$ for $-4 \leq x \leq 12$ The point P on the curve has x coordinate 2(a) (i) Use the graph to find an estimate for the gradient of the curve at P
(3)(ii) Hence find an equation of the tangent to the curve at P .Give your answer in the form $y = mx + c$
(2)The equation $f(x) = k$ has exactly two different solutions for $-4 \leq x \leq 12$ (b) Use the graph to find the two possible values of k ,
(2)**(Total for Question 18 is 7 marks)**

PAST PAPER SOLUTION

Q#: 18

Marks: 7

TOPIC: **Estimating Gradients**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

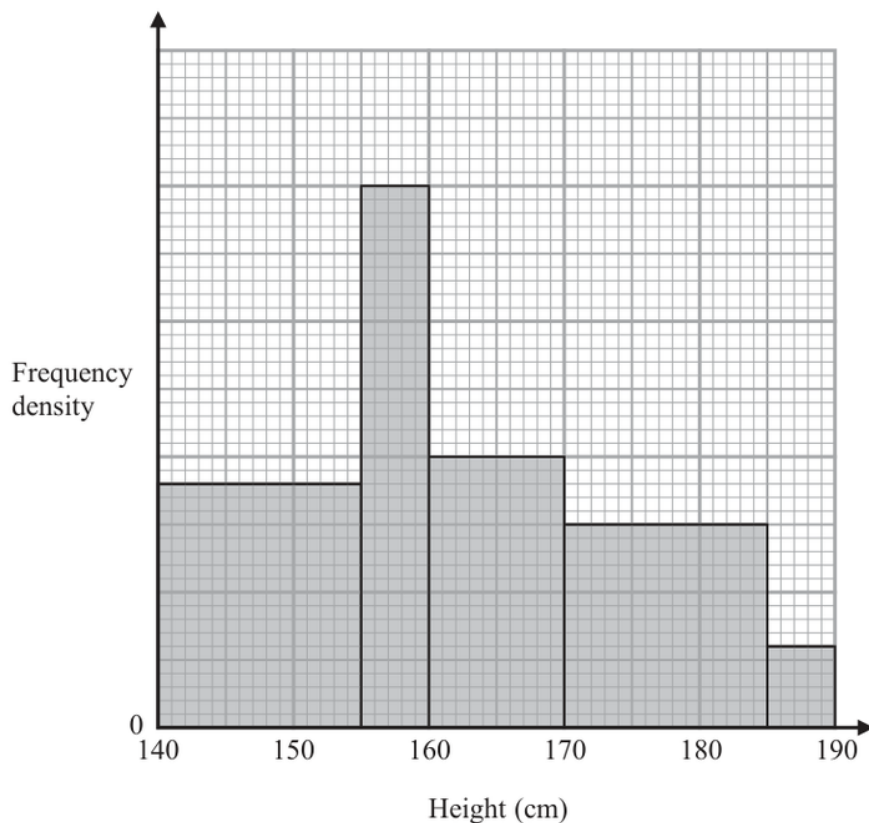
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No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19** **Marks: 4****TOPIC: Histograms**

Jan 2020 P1HR - Jan 2020 | P1HR

19

The histogram gives information about the heights of all the Year 11 students at a school.

There are 160 students in Year 11 with a height between 155 cm and 170 cm.

Work out the total number of students in Year 11 at the school.

(Total for Question 19 is 4 marks)

PAST PAPER SOLUTION

Q#: 19

Marks: 4

TOPIC: **Histograms**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

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No worked solution is saved yet.

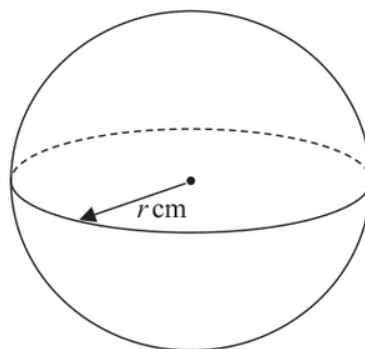
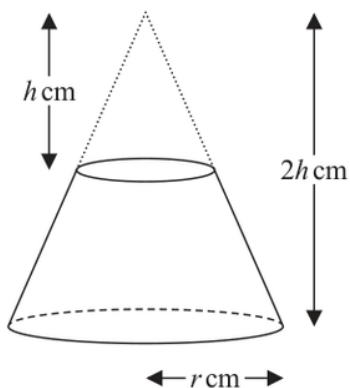
EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Area & Volume of Similar Shapes**

Jan 2020 P1HR - Jan 2020 | P1HR

20 The diagram shows a frustum of a cone and a sphere.

The frustum is made by removing a small cone from a large cone.
The cones are similar.

Diagram **NOT** accurately drawnThe height of the small cone is h cm.The height of the large cone is $2h$ cm.The radius of the base of the large cone is r cm.The radius of the sphere is r cm.

Given that the volume of the frustum is equal to the volume of the sphere,

find an expression for r in terms of h .

Give your expression in its simplest form.

 $r = \dots\dots\dots$ **(Total for Question 20 is 5 marks)**

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

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No worked solution is saved yet.

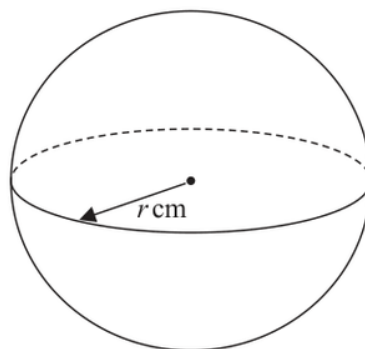
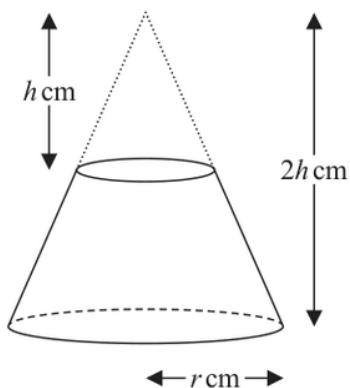
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Give your expression in its simplest form.

 $r = \dots\dots\dots$ **(Total for Question 20 is 5 marks)**

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Area & Volume of Similar Shapes**

Jan 2020 P1HR - Jan 2020 | P1HR

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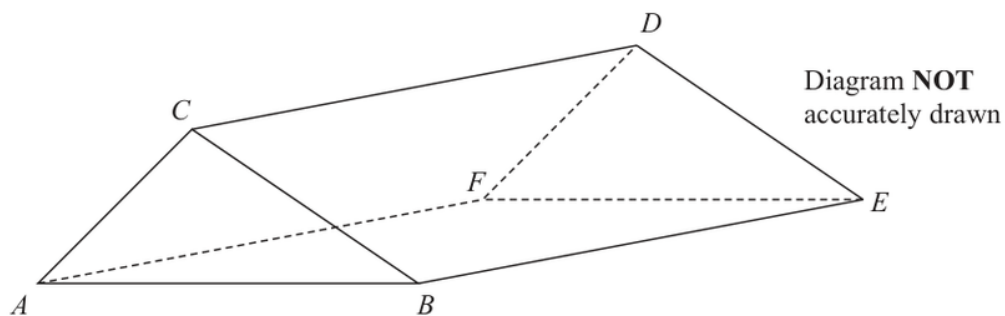
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21** **Marks: 6****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

- 21** The diagram shows the prism $ABCDEF$ with cross section triangle ABC .



Angle $BEC = 40^\circ$ and angle ACB is obtuse.
 $AC = 6$ cm and $CE = 13$ cm

The area of triangle ABC is 22 cm^2

Calculate the length of AB .

Give your answer correct to one decimal place.

..... cm

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

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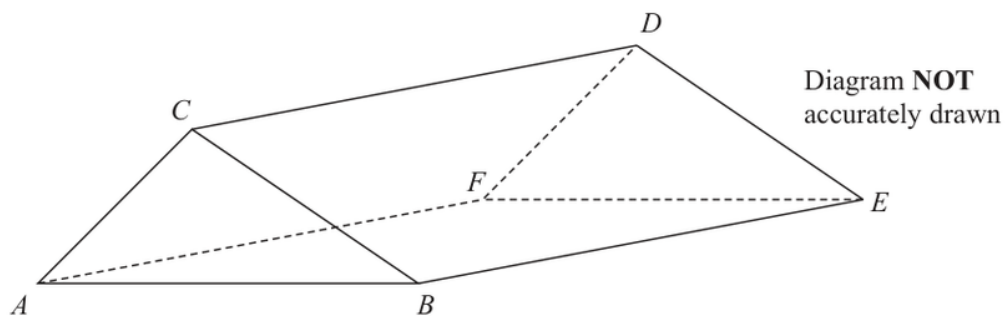
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EA

PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

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Angle $BEC = 40^\circ$ and angle ACB is obtuse.
 $AC = 6$ cm and $CE = 13$ cm

The area of triangle ABC is 22 cm^2

Calculate the length of AB .

Give your answer correct to one decimal place.

..... cm

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION

Q#: 21

Marks: 6

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jan 2020 P1HR - Jan 2020 | P1HR

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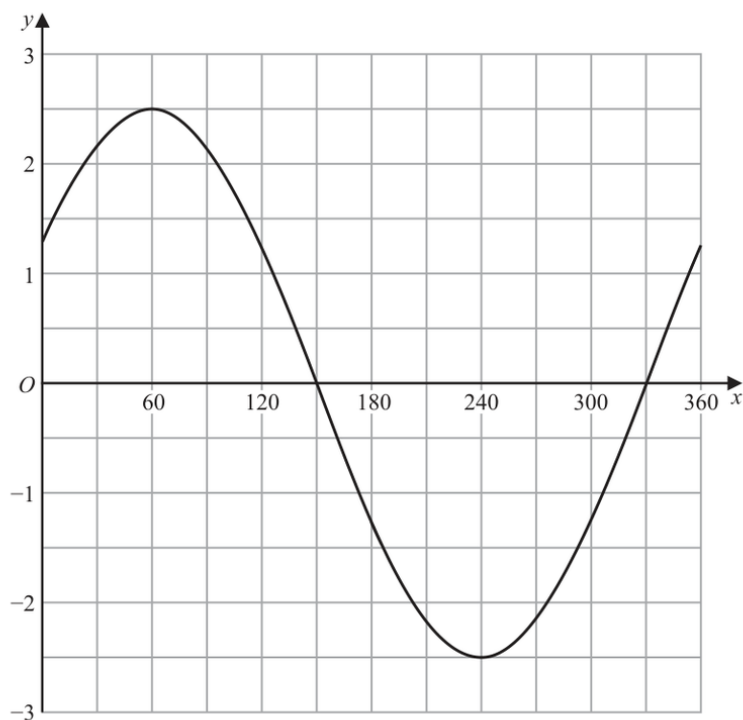
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EA

PAST PAPER QUESTION**Q#: 22** **Marks: 4****TOPIC: Transformations of Graphs**

Jan 2020 P1HR - Jan 2020 | P1HR

22 The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid.



(a) Find the value of a and the value of b .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(2)

Another curve C has equation $y = f(x)$

The coordinates of the minimum point of C are $(4, 5)$

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = f(2x)$

(.....,)

(ii) $y = f(x) - 7$

(.....,)

(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Transformations of Graphs**

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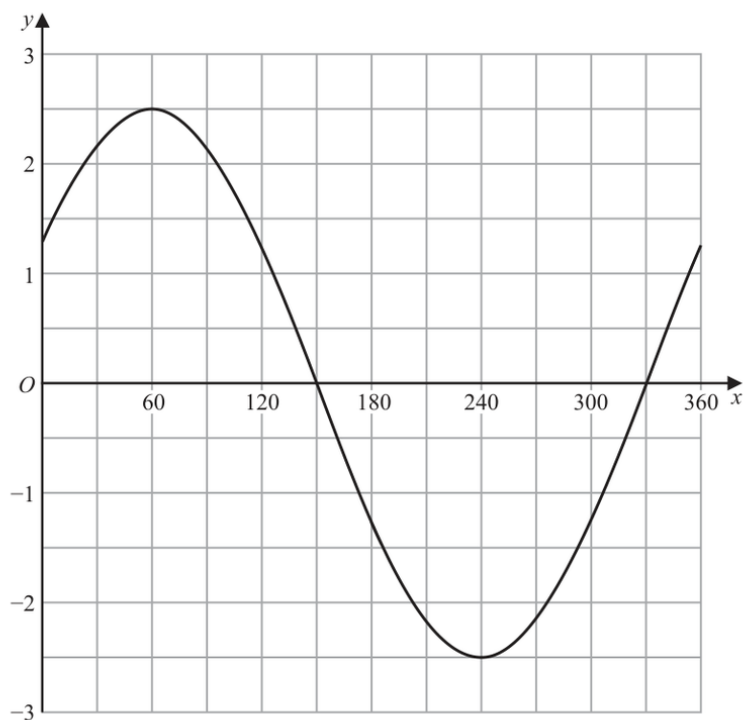
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(2)

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22

Marks: 4

TOPIC: **Transformations of Graphs**

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No worked solution is saved yet.

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PAST PAPER QUESTION**Q#: 23****Marks: 6****TOPIC: Differentiation**

Jan 2020 P1HR - Jan 2020 | P1HR

23 A particle moves along a straight line.The fixed point O lies on this line.The displacement of the particle from O at time t seconds, $t \geq 0$, is s metres where

$$s = t^3 + 4t^2 - 5t + 7$$

At time T seconds the velocity of P is V m/s where $V \geq -5$ Find an expression for T in terms of V .Give your expression in the form $\frac{-4 + \sqrt{k + mV}}{3}$ where k and m are integers to be found. $T = \dots\dots\dots$ **(Total for Question 23 is 6 marks)**

EA

PAST PAPER SOLUTION

Q#: 23

Marks: 6

TOPIC: **Differentiation**

Jan 2020 P1HR - Jan 2020 | P1HR

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$$s = t^3 + 4t^2 - 5t + 7$$

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EA

PAST PAPER SOLUTION

Q#: 23

Marks: 6

TOPIC: **Differentiation**

Jan 2020 P1HR - Jan 2020 | P1HR

WORKED SOLUTION

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No worked solution is saved yet.

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